



Original Research

Ruthenium-106 brachytherapy for choroidal melanoma without transpupillary thermotherapy: Similar efficacy with improved visual outcome



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Treatment outcome

Abstract Purpose: To evaluate efficacy and toxicity of two different protocols for eye-conserving treatment of patients with small to intermediate-sized choroidal melanomas; the current ruthenium-106 (Ru106) brachytherapy protocol and the preceding protocol of Ru106-brachytherapy with transpupillary thermotherapy (Ru106/TTT).

Methods and materials: Long-term outcomes of 449 consecutive patients, of whom 196 (43.6%) treated using Ru106/TTT and 253 (56.3%) treated using Ru106, were compared in terms of local control, survival, eye preservation and visual outcome.

Results: Median follow-up was 82.8 months. Patients in the Ru106-group had smaller, less centrally located tumours and better pre-treatment visual acuity (VA). Five-year cumulative incidence of local failure was 11.2% for Ru106/TTT and 5.2% for Ru106, which was not statistically significant after correction for differences in baseline characteristics (hazard ratio for Ru106 = 0.57, $p = 0.14$). Cumulative incidence of distant metastases was 11.2 versus 6.2%, and cumulative incidence of cause-specific death was 22.4 versus 5.5% for Ru106/TTT and Ru106 respectively. Enucleation was performed in 9.2 versus 4.0% for Ru106/TTT versus Ru106; 5.1 versus 3.2% for local failure and 2.6 versus 0.8% for complications. At one year of follow-up, significantly more patients had lost useful vision ($VA < 0.33$) in the Ru106/TTT-

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group than in the Ru106-group (50.0 versus 24.5%). After two and three years, the differences decreased (54.6 versus 34.0% and 61.7 versus 45.8%, respectively) and lost statistical significance. **Conclusions:** Both the current Ru106 and the preceding Ru106/TTT-protocols provided excellent tumour control, cosmetic and functional eye preservation and vital prognosis. The Ru106-protocol yielded prolonged preservation of VA and should be regarded the current standard of treatment. © 2016 Elsevier Ltd. All rights reserved.

1. Introduction

Choroidal melanoma is the most frequently diagnosed malignant tumour of the eye, with an incidence of about 1 per 100,000 in the Netherlands [1]. Leiden University Medical Center (LUMC) is national referral centre for eye-conserving treatment of choroidal melanoma. Patients with small to intermediate-sized melanomas have been treated with ruthenium-106 brachytherapy (Ru106) since 1984. In 1995, transpupillary thermotherapy (TTT) was introduced as an additive to the treatment of choroidal melanomas [2–4]. Treatment protocols combining Ru106-brachytherapy and TTT (Ru106/TTT), the so-called sandwich therapy, were designed to preserve visual acuity (VA) by reducing brachytherapy dose while maximising local control by TTT [4–6]. This combined Ru106/TTT therapy appeared to yield a high probability of tumour control and eye preservation [7,8]. However, loss of VA was not significantly reduced compared to treatment with Ru106 alone [7,8].

To improve preservation of VA and maintain high local control, treatment protocol was again changed in 2008. Hence, patients were treated with Ru106 only and brachytherapy doses were specified to the tumour apex, instead of the scleral surface, to obtain a more tumour size-specific treatment [8]. With this protocol, use of TTT was limited to patients in whom insufficient tumour regression was observed after brachytherapy.

The aim of the current analyses was to evaluate the performance of the current ‘Ru106’ protocol and the preceding ‘Ru106/TTT’ protocol. For this, long-term outcomes of two consecutive treatment cohorts, from 2004 to 2008 (Ru106/TTT) and 2008 to 2011 (Ru106), were compared in terms of local control, metastasis, eye preservation, VA and treatment complications.

2. Materials and methods

2.1. Patients and treatment

All choroidal melanoma patients treated with Ru106-brachytherapy at LUMC from January 2004 to June 2011 were included. All patients were prospectively registered in a clinical database. Patients treated for iris and ciliary body melanomas were excluded and will be reported separately [9].

Diagnosis of choroidal melanoma was based on ophthalmoscopic findings, ultrasonography (USG) and

fluorescence angiography (FAG). At the time of diagnosis, patients were screened for distant metastasis using chest radiography, USG of the liver and routine blood tests.

Patients with small to intermediate-sized choroidal melanomas with tumour thickness ≤ 8 mm and basal diameters ≤ 16 mm, were considered eligible for Ru106-brachytherapy. Tumour size (largest basal diameter) and thickness (distance of the scleral surface to the tumour apex) were measured using USG.

Four different Ru106 applicators (Bebig, Germany) were used: three round applicators (CCA, CCD and CCB) with diameters of 15.3, 17.9 and 20.2 mm and one (COB) with a groove for the optic nerve.

Patients were treated according to the treatment protocols presented in Table 1. The Ru106/TTT-protocol was used until 1st March 2008 [8]. Brachytherapy dose was specified at the scleral surface and varied from 400 Gy to 800 Gy by tumour localisation and tumour thickness. Brachytherapy was followed by at least one TTT session within 2 months for all centrally located tumours. In case a peripherally located tumour was accessible for TTT, the protocol for central tumours was used. Additional TTT was allowed (and often used) in this protocol, when tumour regression was judged too slow. From 1st March 2008 onwards, patients were treated according to the current Ru106-protocol. For all tumours, brachytherapy dose was 130 Gy, specified to the tumour apex, with a maximum scleral dose of 1000 Gy. In selected cases with tumour thickness > 7 mm scleral doses > 1000 Gy were allowed. To limit reduction of brachytherapy dose compared with the previous protocol, a minimum scleral surface dose of 300 Gy was used. In principle, TTT was not included in the initial treatment, but only used in case of insufficient tumour regression with signs of active residual tumour. However, a limited number of patients received TTT as part of the initial treatment during the transition phase in 2008 (Table 1).

Brachytherapy dose rate was standardised to 100 Gy per 24 h, by correcting the prescribed dose by a factor equal to $(\text{standard dose rate}/\text{actual dose rate})^{0.2}$, adapted from Ellis *et al.* [10] In practice, this resulted in a dose correction of 2–10% [8].

2.2. Outcomes

Primary outcome for this analysis was local tumour control. Secondary outcomes were distant metastases

Table 1
Treatment protocols and observed treatment by period.

Period	Protocol					Treatment			
	Tumour localisation	Tumour thickness	Dose prescribed	Dose specification	TTT prescribed	No. of patients	Scleral dose equivalent (Gy)	Tumour top dose equivalent (Gy)	TTT included in initial treatment
		mm	Gy			N	Mean (range)	Mean (range)	N (%)
Ru106/TTT ('04–'07)	Central	≤5	400	Scleral surface	Yes	67	416.8 (353.8–660.0)	153.6 (75.5–367.4)	64 (95.5)
	Central	>5	600	Scleral surface	Yes	38	599.6 (497.1–690.0)	88.6 (27.0–156.6)	38 (100.0)
	Peripheral*	≤3	600	Scleral surface	No	11	437.2 (387.4–611.4)	213.5 (163.4–284.4)	9 (81.8)
	Peripheral*	>3	800	Scleral surface	No	80	571.1 (376.0–835.7)	138.2 (26.3–387.4)	57 (71.3)
Total Ru106/TTT	All	All	400–800	Scleral surface	Yes/no	196	516.4 (353.8–835.7)	138.1 (26.3–387.4)	168 (85.7)
Ru106 ('08–'11)	All	All	130	Tumour apex	No	253	449.3 (247.4–1159.5)	142.9 (41.5–345.4)	27 (10.7)

Ru106 = ruthenium-106 brachytherapy; TTT = transpupillary thermotherapy.

* If not accessible to TTT.

and death, eye preservation, treatment-related complications and VA after treatment.

After initial treatment, all patients were followed for at least five years. In some patients with complete remission after two years, follow-up was continued by their local ophthalmologist, who provided follow-up information. Vital status of all patients was obtained from hospital records or oncological registration. Causes of death were collected from hospital records, oncological registration or from the local ophthalmologist or general practitioner.

Tumour response was evaluated by fundoscopy and USG. In case of residual thickness at the tumour site, FAG was performed to differentiate between non-active residual changes and active residual tumour. Local tumour control was defined as a flat scar, or a regressed lesion not showing any signs of tumour activity. Local failure was defined as: (1) insufficient tumour regression on USG with signs of tumour activity on FAG necessitating additional treatment, or (2) documented tumour growth, or (3) true recurrence (growth after initial tumour regression). This definition differs from our previous study [8], wherein insufficient tumour regression necessitating additional treatment by TTT was incorporated in the treatment protocol and not considered a local failure.

Metastases were detected by routine follow-up investigations (liver USG and blood chemistry tests) which were performed twice yearly, or by evaluation of symptomatic patients.

Diplopia was recorded whenever patients complained about double vision and an orthoptic evaluation was done, even if complaints subsided with time or exercise. VA was measured using Snellen charts. Acute and late treatment-related complications were evaluated by fundoscopy at each follow-up visit. FAG and optical coherence tomography were performed when indicated.

2.3. Statistics

Differences between nominal variables were calculated by chi-square test or Fisher's exact test. Differences between categorical variables were calculated using

Mann–Whitney U test. The difference between continuous variables was calculated using the independent samples median test. Median follow-up was estimated by using reverse Kaplan–Meier's methodology [11].

Time to local failure, metastasis and death were calculated from the first day of brachytherapy to the date of the occurrence of the event with censoring at the date of last follow-up for patients alive and event-free. A competing risk model [12] was used to analyse local failure, distant metastasis and death by treatment protocol. Differences between cumulative incidence for each event failure were tested using Gray's test [13]. To investigate risk factors associated with cumulative incidence of local failure and metastasis Fine and Gray's regression model was used to perform univariate and multivariate analysis subsequently.

For analyses of eye preservation, retreatments, treatment complications, only data of the first five years after treatment were used to prevent distortion of results by the difference in follow-up time of the two consecutive cohorts. To determine the effect of treatment protocol on eye preservation and loss of useful vision (according to the World Health Organisation [WHO] classification defined as VA less than 0.33) a Cox proportional hazards model was used to perform univariate and multivariate regression analysis subsequently. All clinical characteristics which were significantly different between the groups were entered in the univariate model. Covariates with a p-value < 0.10 in univariate analysis were entered in the multivariate model.

Statistical analyses were performed using SPSS software (version 23.0) and R (version 2–15.3). Analysis concerning competing risks was performed by using the 'mstate' and 'cmprsk' library [14,15]. For all analyses, a two-sided p-value < 0.05 was considered statistically significant.

3. Results

A total of 474 patients were treated from January 2004 to July 2011. Due to missing follow-up information, 25

were excluded, leaving 449 patients for analyses. 196 Patients (43.7%) were treated according to the Ru106/TTT-protocol and from March 2008 onwards 253 (56.3%) patients were treated according to the Ru106-protocol (Table 1).

Median follow-up was 82.8 months (95% confidence interval [95% CI]: 77.9–87.6); 114.6 months (95% CI: 110.8–118.4) for the Ru106/TTT-group and 68.4 months (95% CI: 66.0–70.8) for the Ru106-group.

Table 1 presents an overview of both treatment protocols, as well as the treatment patients actually received. Brachytherapy dose at the sclera was significantly higher ($p < 0.0001$) in the Ru106/TTT-group compared with the Ru106-group. However, there was no statistically significant difference ($p = 0.31$) in tumour thickness dose between the groups. TTT was

included in the initial treatment significantly more often ($p < 0.0001$) in the Ru106/TTT-group.

Patient and tumour characteristics are summarised in Table 2. Patients in the Ru106-group had significantly smaller, less centrally located tumours, and better VA before treatment.

3.1. Local failure, metastasis and death

Cumulative incidence of local failure and metastasis for both treatment groups are presented in Fig. 1. The cumulative incidence of local failure was lower (sub-distribution hazard ratio (HR_s) = 0.55, 95% CI = 0.29–1.05), $p = 0.069$ for patients who were treated according to the current Ru106-protocol than for patients who were treated according to the preceding

Table 2
Patient and tumour characteristics at baseline.

Characteristics	Ru106/TTT N (%)	Ru106 N (%)	p-Value
Sex			
Male	96 (49.0)	116 (45.8)	0.51
Female	100 (51.0)	137 (54.2)	
Age, mean (SD)	61.4 (12.8)	63.5 (12.0)	0.07
Diabetes mellitus	29 (14.8)	38 (15.0)	0.95
Eye involved			
Right	105 (53.6)	131 (51.8)	0.71
Left	91 (46.4)	122 (48.2)	
Vision ipsilateral eye, median (IQR)	0.70 (0.40–0.90)	0.90 (0.55–1.00)	<0.0001
≤0.10	17 (8.7)	7 (2.8)	0.006
0.11–0.33	26 (13.3)	22 (8.7)	0.12
0.33–0.50	35 (17.9)	34 (13.4)	0.20
0.51–0.80	63 (32.1)	62 (24.5)	0.074
>0.80	55 (28.1)	128 (50.6)	<0.0001
Tumour pigmentation			
Melanotic	146 (74.5)	184 (72.7)	0.67
Amelanotic	24 (12.2)	26 (10.3)	0.51
Partly amelanotic	26 (13.3)	43 (17.0)	0.27
Tumour localisation			
Central	76 (38.8)	66 (26.1)	0.004
Juxta papillary	29 (14.8)	24 (9.5)	0.085
Mid peripheral	53 (27.0)	113 (44.7)	<0.0001
Peripheral	38 (19.4)	50 (19.8)	0.92
Tumour basal diameter – median (IQR)	11.0 (9.5–13.2)	10.6 (9.1–12.6)	0.078
3–6 mm	1 (0.5)	0 (0.0)	0.44
6–9 mm	29 (14.8)	54 (21.3)	0.074
9–12 mm	86 (43.9)	109 (43.1)	0.87
12–15 mm	71 (36.2)	74 (29.2)	0.12
15–18 mm	9 (4.6)	15 (5.9)	0.53
18 mm or larger	0 (0.0)	1 (0.4)	1.00
Tumour thickness, median (IQR)	4.3 (3.4–5.8)	3.6 (3.0–4.7)	<0.0001
3 mm or smaller	23 (11.7)	60 (23.7)	0.001
3–6.0 mm	128 (65.3)	172 (68.0)	0.55
6–9 mm	43 (21.9)	21 (8.3)	<0.0001
9–12 mm	2 (1.0)	0 (0.0)	0.19
TNM stage			0.003
I	43 (21.9)	82 (32.4)	0.013
IIA	120 (61.2)	147 (58.1)	0.50
IIB	33 (16.8)	24 (9.5)	0.021
Total	196 (100.0)	253 (100.0)	

Ru106 = ruthenium-106 brachytherapy; TTT = transpupillary thermotherapy; SD = standard deviation; IQR = interquartile range; TNM = tumour-node-metastasis.

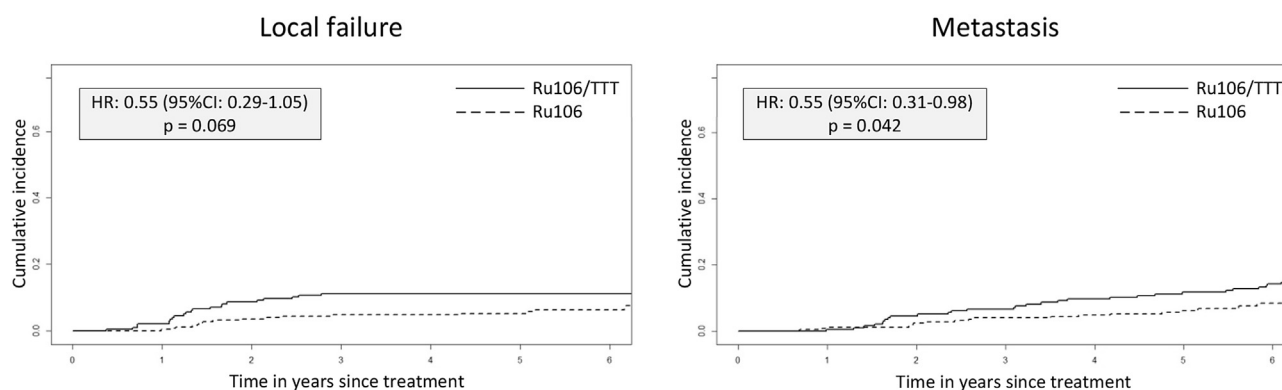


Fig. 1. Cumulative incidence of local failure and metastasis by treatment group. Ru106 = ruthenium-106 brachytherapy; TTT = transpupillary thermotherapy; HR = hazard ratio; 95% CI = 95% confidence interval.

Ru106/TTT-protocol. At five years after treatment, cumulative incidence of local failure was 11.2% (95% CI = 6.8–15.6%) in the Ru106/TTT-group versus 5.2% (95% CI = 2.4–7.9%) in the Ru106-group.

Metastasis of choroidal melanoma was diagnosed significantly less often in the Ru106-group ($HR_s = 0.55$, 95% CI = 0.31–0.98, $p = 0.042$). Five-year cumulative incidence of metastasis were 11.2% (95% CI = 6.8–15.6%) and 6.2% (95% CI = 3.1–9.2%) in the Ru106/TTT and Ru106-groups respectively.

However, after correction for differences in tumour size, localisation and stage between the groups, no statistically significant difference for local failures and metastasis between the groups was observed (Table 3). Tumour diameter appeared to be an independent risk factor for local failure ($p < 0.001$). A trend towards significance was observed for higher tumour stage and more central localisation. For metastasis, tumour thickness was a nearly significant risk factor ($p = 0.065$).

In the first five years after treatment, 106 (23.6%) patients died; 73 (37.2%) in the Ru106/TTT-group and 33 (13.0%) in the Ru106-group. Death due to choroidal melanoma was the most common cause of death; 44 (60.3%) cases in the Ru106/TTT-group and 14 (42.4%) in the Ru106 group. Death due to other malignancies was reported in 8 (11.0%) and 5 (15.2%) patients of the Ru106/TTT and the Ru106-group respectively.

Overall, no statistically significant difference in the number of retreatments was observed between the groups (Ru106/TTT 11.2 versus Ru106 6.7%, $p = 0.09$). When excluding retreatment with TTT, 14 (7.1%) patients in the Ru106/TTT-group and 13 (5.1%) in the Ru106-group received any retreatment ($p = 0.38$).

3.2. Retreatments

Table 4 presents retreatments for local failure at 5 years after treatment. Overall, no statistically significant difference in the number of retreatments was observed between the groups (Ru106/TTT 11.2 versus Ru106 6.7%). Also when retreatment with TTT was excluded;

14 (7.1%) patients in the Ru106/TTT-group and 13 (5.1%) in the Ru106-group received any retreatment.

Enucleation was the most commonly performed retreatment for local failure in both groups; 10 (5.1%) of the patients in the Ru106/TTT-group and 8 (3.2%) of the patients in the Ru106-group underwent enucleation for local failure. Retreatment using one or more TTT sessions was more common in the Ru106/TTT-group (4.1%) compared with the Ru106-group (1.6%), while retreatment using Ru106-brachytherapy was more common in the Ru106-group (1.6 versus 0.5%). However, none of the observed differences reached statistical significance.

3.3. Eye preservation and ocular treatment complications

At five years after primary treatment, enucleation (both for local failure and treatment complications) was performed in 18 patients (9.2%) of the Ru106/TTT-group and 10 patients of the Ru106-group (4.0%). Univariate Cox regression analyses showed that treatment according to the Ru106/TTT-protocol, younger age, worse pre-treatment VA, larger tumour diameter and thickness and a higher tumour-node-metastasis stage, were risk factors for enucleation. Multivariate Cox regression analyses showed that only younger age ($HR: 0.96$, 95% CI: 0.93–0.99, $p = 0.008$) and larger tumour thickness ($HR: 1.46$, 95% CI: 1.18–1.81, $p = 0.001$) were independent risk factors for enucleation.

An overview of treatment complications is presented in Table 5. At five years after treatment, ocular complications had occurred in 46.4% of the patients of the Ru106/TTT-group and in 44.7% of the Ru106-group ($p =$ not significant). No statistically significant difference in acute complications (Ru106/TTT $n = 60$, 30.6% versus Ru106 $n = 68$, 26.9%) and late complications (Ru106/TTT $n = 46$, 23.5% versus Ru106 $n = 71$, 28.1%) was found. Five patients (2.6%) in the Ru106/TTT-group underwent enucleation for treatment complications, compared with 2 (0.8%) in the Ru106-group ($p = 0.25$).

Table 3
Multivariate Fine and Gray competing risk analyses for local failure and metastasis.

Predictor	Local failure			Metastasis		
	HR _s	95% CI	p-value	HR _s	95% CI	p-value
Treatment protocol ^a	0.57	0.27–1.20	0.14	0.71	0.37–1.38	0.31
Age ^b	0.97	0.95–0.99	0.010	0.99	0.97–1.02	0.56
Gender ^c	1.00	0.53–1.91	0.99	1.42	0.80–2.50	0.23
Tumour diameter ^d	1.47	1.19–1.82	<0.0001	1.14	0.96–1.35	0.14
Tumour thickness ^d	1.04	0.78–1.41	0.78	1.25	0.99–1.58	0.065
Tumour localisation						
Central	Reference category			Reference category		
Juxta papillary	1.35	0.54–3.33	0.52	0.77	0.26–2.24	0.63
Mid peripheral	0.99	0.42–2.33	0.98	1.04	0.54–2.03	0.90
Peripheral	0.29	0.09–1.00	0.05	1.52	0.73–3.19	0.27
TNM stage						
I	Reference category			Reference category		
IIA	0.53	0.19–1.52	0.24	1.30	0.40–4.28	0.66
IIB	0.18	0.03–1.04	0.055	1.80	0.34–9.66	0.49

HR_s = sub-distribution hazard ratio; 95% CI = 95% confidence interval; TNM = tumour-node-metastasis.

^a Ru106/TTT protocol as reference.

^b Age in years.

^c Males as reference.

^d Size in mm.

3.4. Visual outcomes

Table 6 presents the effect of treatment on loss of useful vision (WHO definition <0.33). One year after treatment, 50% of the subjects in the Ru106/TTT-group had lost useful vision, compared with 24.5% of the subjects in the Ru106-group. This difference was statistically significant, even after correction for the differences in baseline characteristics between the groups (HR 0.66, 95% CI: 0.47–0.91, $p = 0.012$). During follow-up, the difference between the groups decreased from 54.6 versus 34.0% after two years, to 61.7 versus 45.8% after three years for the Ru106/TTT and the Ru106-groups, respectively. The differences in loss of useful vision at 2 and 3 years of follow-up did not reach statistical significance in multivariate analyses.

Independent risk factors for loss of useful vision at three years of follow-up were: pre-treatment VA (HR 0.38, 95%

Table 4
Retreatments within 5 years after treatment.

Retreatment	Ru106/TTT	Ru106
	N (%)	N (%)
Any retreatment	22 (11.2)	17 (6.7)
Ruthenium106	1 (0.5)	4 (1.6)
Enucleation	10 (5.1)	8 (3.2)
TTT	6 (3.1)	3 (1.2)
TTT followed by TTT	2 (1.0)	1 (0.4)
TTT followed by enucleation	3 (1.5)	0 (0.0)
Ru106 followed by TTT	0 (0.0)	1 (0.4)
Total	196 (100.0)	253 (100.0)

Ru106 = ruthenium-106 brachytherapy; TTT = transpupillary thermotherapy.

Table 5
Treatment complications within 5 years after treatment.

Complication	Ru106/TTT	Ru106
	N (%)	N (%)
Retinal ablation	28 (14.3)	10 (4.0)
Uveitis	4 (2.0)	4 (1.6)
Vascular accident	17 (8.7)	2 (0.8)
Diplopia	24 (12.2)	54 (21.3)
Retinopathy	46 (23.5)	73 (28.9)
Opticopathy	2 (1.0)	5 (2.0)
Neovascular glaucoma	1 (0.5)	0 (0.0)
Total patients	196 (100.0)	253 (100.0)

Ru106 = ruthenium-106 brachytherapy; TTT = transpupillary thermotherapy.

CI: 0.26–0.57, $p < 0.0001$), central tumour location (HR 0.35, 95% CI: 0.22–0.58, $p < 0.0001$), and tumour thickness (HR 1.13, 95% CI: 1.04–1.22, $p < 0.0001$).

4. Discussion

This study showed that the Ru106 and the Ru106/TTT treatment protocols yielded excellent local control. Patients treated according to the Ru106-protocol had a 96% eye preservation rate (at five years after treatment) and significantly longer preservation of VA. Main differences between these two protocols were use of TTT (standard for central tumours in the Ru106/TTT-protocol and not used in the Ru106-protocol) and radiotherapy dose (400–800 Gy specified at the scleral surface in the Ru106/TTT-protocol and 130 Gy specified at the tumour apex in the Ru106-protocol).

Local failure after treatment for choroidal melanoma had occurred in 11.2% of the patients in the Ru106/TTT-group and in 5.2% of the patients in the Ru106-group at five years after treatment. The latter is in the range of our previously reported five-year local recurrence rates (3.9% and 6.1%) [7,8]. Since the definition of local failure was more stringent in the current study (additional TTT for slow tumour regression was considered part of the protocol in previous studies, but is now considered a local failure) local control rates may even be higher using our current Ru106-protocol [7,8]. The probability of local failure of 5.2% after five years compares favourably to rates reported by other groups using Ru106 (2.1–27%) [5,16–24].

Patients in the current treatment group (Ru106) had smaller, more peripherally localised tumours compared with the preceding treatment group (Ru106/TTT); a trend which was also observed in the Collaborative Ocular Melanoma Study [25]. After correcting for this risk factor no statistically significant difference in the cumulative incidence of local failure was observed between the treatment groups (HR = 0.57, 95% CI = 0.27–1.20).

Within five years after therapy, enucleation was performed in 4.0% of patients in the Ru106-group and in

Table 6

Effect of treatment protocol on loss of useful vision during follow-up.

Effect of treatment protocol ^a	Ru106/TTT	Ru106	Univariate analyses			Multivariate analyses ^b		
	VA < 0.33 N (%)	VA < 0.33 N (%)	HR	95% CI	p-value	HR	95% CI	p-value
At 1-year follow-up	98 (50.0)	62 (24.5)	0.42	0.31–0.58	<0.0001	0.66	0.47–0.91	0.012
At 2-year follow-up	107 (54.6)	86 (34.0)	0.53	0.40–0.71	<0.0001	0.84	0.62–1.15	0.28
At 3-year follow-up	121 (61.7)	116 (45.8)	0.63	0.49–0.82	<0.0001	0.93	0.71–1.23	0.62

Ru106 = ruthenium-106, TTT = transpupillary thermotherapy, VA = visual acuity, HR = hazard ratio, 95% CI = 95% confidence interval.

^a The Ru106/TTT treatment protocol is the reference group.^b Correction for covariates with p-value < 0.10 in univariate analyses, which were for all analyses: pre-treatment vision, location tumour, diameter, thickness and TNM stage.

9.2% in the Ru106/TTT-group. After correction for the differences between the two treatments groups, no statistically significant difference in 5-year enucleation rate remained. Our 5-year enucleation rates are in the low range of other studies using ruthenium brachytherapy (2.0–28%) [5,17,18,21,22,24,26,27] and substantially lower than the enucleation rate of 16% in a study using stereotactic radiotherapy [28].

Ocular complications at five years after treatment had occurred in 46.4% in the Ru106/TTT-group and in 44.7% in the Ru106-group, which is substantially better than in our previous report (72.2%) [8] and other reports (57–89.9%) [24,27]. The higher incidence of vascular incidents (8.7 versus 0.8%) and retinal detachments (14.3 versus 4.0%) in the Ru106/TTT-group is probably related to use of TTT [29,30]. The higher incidence of diplopia in the Ru106-group (21.3 versus 12.2%) was unexpected. This may be caused by better VA with subsequently more symptoms of diplopia, and warrants further investigation.

One year after treatment, loss of useful vision had occurred significantly more often in the Ru106/TTT group (50.0%) than in the Ru106-group (24.5%). This may be explained by the immediate effect of TTT on VA, due to direct damage and complications such as vascular accidents and serous retinal detachments [29,30]. One other study compared VA between patients treated using Ru106 and Ru106 combined with TTT; a decline in VA was observed in 71% in the Ru106/TTT-group versus 64% in the Ru106 group, although no statistical significance was reached [24].

After 2 and 3 years of follow-up, the differences in VA decreased between the two groups and did not reach statistical significance anymore. Hence, it appears that tumour location and factors related to radiotherapy determine long-term preservation or loss of VA. In our study, patients with thicker tumours (and thus receiving higher radiotherapy doses) and more centrally located tumours had significantly worse visual outcomes. Pre-treatment VA [31], central localisation of the tumour [32,33] and tumour thickness [32] were also identified in other studies as independent risk factors for vision loss.

At five years after treatment, distant metastases were diagnosed in 11.2% of the patients in the Ru106/TTT-group and in 6.2% of the patients in the Ru106-group.

After correction for the larger and more advanced stage of the tumours, no significant difference was found (HR = 0.72, 95% CI = 0.37–1.38). Five-year cumulative incidence of metastasis was in the lower range of other reported incidences (5–21%) [16,18,19,22,24,27].

As expected, metastasised choroidal melanoma was the most common cause of death in both treatment groups; 22.4% of the patients in the Ru106/TTT-group and 5.5% of the Ru106-group at five years after treatment. The incidence in the Ru106-group was lower than our previously reported (10.8%) [8] and other reported incidences (5–12.3%); possibly due to smaller tumour size [16,18,27].

In conclusion, the current Ru106 and preceding Ru106/TTT-protocols for the treatment of choroidal melanomas provided similar excellent tumour control, cosmetic and functional eye preservation and vital prognosis. The Ru106-protocol, however, yielded a prolonged preservation of VA, and should be regarded the current standard of treatment.

Conflict of interest statement

None declared.

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